

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

**Product Name** Fluoride Standard - 0.1 M NaF  
**Product No** 940906  
**Pure substance/mixture** Mixture

### Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Use as laboratory reagent  
**Uses advised against** No Information available

### **Supplier**

|                                                                                                                                                                                                    |                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Fisher Scientific Korea<br>D5,D6, Incheon Airport Logistics<br>Complex<br>150, Gonghangdong-Ro Mapo-Gu 296<br>Beon-Gil<br>Jung-Gu, Incheon 400-340<br>Tel: +82-2-6196-5500<br>Fax: +82-2-6196-5501 | Thermo Fisher Scientific©<br>Water and Lab Products<br>22 Alpha Road<br>Chelmsford, MA 01824, USA<br>1-978-232-6000 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

### **E-mail address**

Chem.KR@thermofisher.com info.water@thermo.com

### Emergency Telephone Number

**Emergency telephone:** Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

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Based on available data, the classification criteria are not met

## **Environmental hazards**

Based on available data, the classification criteria are not met

## **Label Elements**

## **Other Hazards**

## **NFPA**

Health  
4

Flammability  
0

Instability  
0

Physical hazards  
N/A

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### **3.2. Mixtures**

| Component       | Common Name              | CAS-No    | Index No. | Weight % |
|-----------------|--------------------------|-----------|-----------|----------|
| Water           | No information available | 7732-18-5 | KE-35400  | 99 - 100 |
| Sodium Fluoride | No information available | 7681-49-4 | KE-31540  | 0.5 - 1  |

## **SECTION 4: FIRST AID MEASURES**

### **First aid measures**

#### **General Advice**

Use first aid treatment according to the nature of the injury. Get medical attention immediately if symptoms occur. Show this safety data sheet to the doctor in attendance.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### **Inhalation**

Move to fresh air. Get medical attention immediately if symptoms occur.

#### **Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### **Self-Protection of the First Aider**

No special precautions required.

### **Most important symptoms and effects, both acute and delayed**

#### **Most important symptoms and effects**

None reasonably foreseeable

### **Indication of any immediate medical attention and special treatment needed**

#### **Notes to Physician**

Treat symptomatically

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## SECTION 5: FIREFIGHTING MEASURES

### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

### Unsuitable Extinguishing Media

No information available

### Specific Hazards Arising from the Chemical

No information available.

### Explosion Data

**Sensitivity to Mechanical Impact** None

**Sensitivity to Static Discharge** None

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment.

#### Environmental Precautions

Should not be released into the environment. See Section 12 for additional ecological information.

### Methods and Material for Containment and Cleaning Up

#### Methods for Containment

Prevent further leakage or spillage if safe to do so.

#### Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

#### Handling

Wear personal protective equipment  
Ensure adequate ventilation  
Avoid contact with skin, eyes and clothing  
Avoid ingestion and inhalation

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage

Keep container tightly closed in a dry and well-ventilated place  
Store at room temperature in the original container  
Keep away from direct sunlight

#### Incompatible Products

No information available

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component       | CAS-No    | Korea                      | ACGIH TLV                  | OSHA PEL                             |
|-----------------|-----------|----------------------------|----------------------------|--------------------------------------|
| Water           | 7732-18-5 | Not listed                 | Not listed                 | Not listed                           |
| Sodium Fluoride | 7681-49-4 | TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> |

| Component | CAS-No | European Union | The United Kingdom | Germany |
|-----------|--------|----------------|--------------------|---------|
|-----------|--------|----------------|--------------------|---------|

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|                 |           |            |                                                                       |                                                                                                                          |
|-----------------|-----------|------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Water           | 7732-18-5 | Not listed | Not listed                                                            | Not listed                                                                                                               |
| Sodium Fluoride | 7681-49-4 | Not listed | STEL: 7.5 mg/m <sup>3</sup> 15 min<br>TWA: 2.5 mg/m <sup>3</sup> 8 hr | TWA: 1 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 8<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden).<br>MAK<br>Haut |

## Biological Exposure Indices

| Component       | CAS-No    | Biological Exposure Indices |
|-----------------|-----------|-----------------------------|
| Water           | 7732-18-5 | Not listed                  |
| Sodium Fluoride | 7681-49-4 | Not listed                  |

## Appropriate engineering controls

**Engineering Measures** None under normal use conditions

## Individual protection measures, such as personal protective equipment

**Personal protective equipment** Use only those certified by the Korea Occupational Safety and Health Administration.

**Eye/face Protection** Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:.  
Face-shield.

**Skin and Body Protection** Wear protective gloves/clothing.

**Respiratory Protection** None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical State | Liquid                   |
| Appearance     | Clear                    |
| Odor           | None                     |
| Odor Threshold | No information available |
| pH             | 6.5                      |
| PH Range       | 5.0 - 8.0                |

| Property                     | Values                   | Remarks • Method |
|------------------------------|--------------------------|------------------|
| Melting point/freezing point | No information available |                  |
| Boiling Point/Range          | ~ 100 °C / 212 °F        |                  |
| Flash Point (High in °C)     | N/A                      |                  |
| Evaporation Rate             | No information available |                  |
| Flammability (solid, gas)    | No information available |                  |
| Flammability Limit in Air    |                          |                  |
| Upper flammability limit:    | No information available |                  |
| Lower flammability limit:    | No information available |                  |
| Vapor pressure               | No information available |                  |
| Vapor Density                | No information available |                  |
| Specific Gravity             | No information available |                  |
| Water Solubility             | soluble                  |                  |
| Solubility in other solvents | No information available |                  |
| Partition coefficient        | No information available |                  |
| Autoignition Temperature     | -                        |                  |
| Decomposition Temperature    | No information available |                  |

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|                      |                          |
|----------------------|--------------------------|
| Kinematic viscosity  | No information available |
| Dynamic viscosity    | No information available |
| Explosive Properties | No information available |
| Oxidizing Properties | No information available |

## Other Information

|                  |                          |
|------------------|--------------------------|
| Softening Point  | No information available |
| Molecular Weight | No information available |
| VOC Content(%)   | No information available |
| Density          | No Information available |
| Bulk Density     | No information available |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

No Information available

Contact with acids liberates very toxic gas

### Chemical Stability

Stable under normal conditions

### Possibility of Hazardous Reactions

None under normal processing

### Conditions to Avoid

Extremes of temperature and direct sunlight.

### Incompatible Materials

No information available

### Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Product Information

### Information on likely routes of exposure

|            |                                                |
|------------|------------------------------------------------|
| Inhalation | Not an expected route of exposure.             |
| Ingestion  | No known effect based on information supplied. |
| Eyes       | Not an expected route of exposure.             |
| Skin       | No known effect based on information supplied. |

### Information on Toxicological Effects

|          |                          |
|----------|--------------------------|
| Symptoms | No information available |
|----------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                      |                           |
|----------------------|---------------------------|
| Sensitization        | No information available  |
| Mutagenic Effects    | No information available  |
| Carcinogenicity      | No information available. |
| Reproductive Effects | No information available  |

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**STOT - single exposure** No information available

**STOT - repeated exposure** No information available  
**Target Organs** None known.

**Aspiration hazard** No information available

## Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

**ATEmix (oral)** 5200 mg/kg  
**ATEmix (dermal)** 17500 mg/kg

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity

0% of the mixture consists of components(s) of unknown hazards to the aquatic environment

| Component                    | Freshwater Algae                                                                                                        | Freshwater Fish                                                                                                                                                                                                                       | Water Flea                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Sodium Fluoride<br>7681-49-4 | EC50: = 850 mg/L, 72h static<br>(Desmodesmus subspicatus)<br>EC50: = 272 mg/L, 96h<br>(Pseudokirchneriella subcapitata) | LC50: = 180 mg/L, 96h semi-static<br>(Pimephales promelas)<br>LC50: 38 - 68 mg/L, 96h static<br>(Oncorhynchus mykiss)<br>LC50: = 830 mg/L, 96h semi-static<br>(Lepomis macrochirus)<br>LC50: > 530 mg/L, 96h (Lepomis<br>macrochirus) | EC50: = 98 mg/L, 48h Static<br>(Daphnia magna)<br>EC50: = 338 mg/L, 48h (Daphnia<br>magna) |

### Persistence and Degradability

### Bioaccumulation/ Accumulation

No information available

### Mobility

Will likely be mobile in the environment due to its water solubility.

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste Disposal Methods** Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated Packaging** Improper disposal or reuse of this container may be dangerous and illegal.

## SECTION 14: TRANSPORT INFORMATION

**Road and Rail Transport** Not Regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated  
**Marine Pollutant** Not Applicable

**Special Precautions for User** No special precautions required

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## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

#### International Inventories

| Component       | CAS-No    | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | AICS |
|-----------------|-----------|----------|------|-----------|-------|-----|------|-------|------|------|
| Water           | 7732-18-5 | KE-35400 | X    | 231-791-2 | X     | X   | -    | X     | -    | X    |
| Sodium Fluoride | 7681-49-4 | KE-31540 | X    | 231-667-8 | X     | X   | -    | X     | X    | X    |

| Component       | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water           | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Sodium Fluoride | 7681-49-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component       | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|-----------------|-----------|----------|------------------------------|---------------------------|
| Water           | 7732-18-5 | Listed   | Not applicable               | Not applicable            |
| Sodium Fluoride | 7681-49-4 | Listed   | Not applicable               | Not applicable            |

#### Korean National Regulations

| Component       | CAS-No    | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|-----------------|-----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Water           | 7732-18-5 | Annex 1 - KE-35400 Exempt (Index No. 25)                            | Not applicable       | Not applicable                              |
| Sodium Fluoride | 7681-49-4 | Annex 1 - KE-31540                                                  | Not applicable       | Listed                                      |

| Component       | CAS-No    | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|-----------------|-----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Water           | 7732-18-5 | Not applicable                         | Not applicable                              | Not applicable                                  |
| Sodium Fluoride | 7681-49-4 | 97-1-381 (>25%)                        | Not applicable                              | Not applicable                                  |

| Component       | CAS-No    | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|-----------------|-----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Water           | 7732-18-5 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |
| Sodium Fluoride | 7681-49-4 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component       | CAS-No    | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|-----------------|-----------|-------------------|------------------------------------|-----------------------------------------------------------|
| Water           | 7732-18-5 | Not applicable    | Not applicable                     | Not applicable                                            |
| Sodium Fluoride | 7681-49-4 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component       | CAS-No    | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|-----------------|-----------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Water           | 7732-18-5 | Not applicable                                               | Not applicable               | Not applicable                         |
| Sodium Fluoride | 7681-49-4 | Not applicable                                               | Not applicable               | Not applicable                         |

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| Component       | CAS-No    | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|-----------------|-----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Water           | 7732-18-5 | Not applicable                       | Not applicable                                     | Not applicable                     |
| Sodium Fluoride | 7681-49-4 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component       | CAS-No    | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|-----------------|-----------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Water           | 7732-18-5 | Not applicable                                              | Not applicable                                 | Not applicable                      |
| Sodium Fluoride | 7681-49-4 | Not applicable                                              | TWA: 2.5 mg/m <sup>3</sup>                     | Not applicable                      |

**National Fire Association - Dangerous Substances** Minimum quantity requiring a permit

| Component       | CAS-No    | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|-----------------|-----------|----------------------------|---------------------------|----------------------------------------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
| Water           | 7732-18-5 | Not applicable             | Not applicable            | Not applicable                                                                   | Not applicable              | Not applicable                     | Not applicable              |
| Sodium Fluoride | 7681-49-4 | Not applicable             | Not applicable            | Not applicable                                                                   | Not applicable              | Not applicable                     | Not applicable              |

## Control Parameters

| Component       | CAS-No    | Korea                      | Biological Exposure Indices |
|-----------------|-----------|----------------------------|-----------------------------|
| Water           | 7732-18-5 | Not listed                 | Not listed                  |
| Sodium Fluoride | 7681-49-4 | TWA: 2.5 mg/m <sup>3</sup> | Not listed                  |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component       | CAS-No    | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-----------------|-----------|----------------------------------|----------------------------|
| Water           | 7732-18-5 | Not applicable                   | Not applicable             |
| Sodium Fluoride | 7681-49-4 | Not applicable                   | Not applicable             |

## CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component       | CAS-No    | CERCLA EHS RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|-----------------|-----------|----------------|--------------------------|-------------------------------|
| Water           | 7732-18-5 | Not applicable | Not applicable           | Not applicable                |
| Sodium Fluoride | 7681-49-4 | Not applicable | 1000 lb                  | Not applicable                |

## CLP Classification - Regulation (EC) No 1272/2008

Based on available data, the classification criteria are not met.

EUH210 - Safety data sheet available on request.

## SECTION 16: OTHER INFORMATION

Prepared By

Regulatory Affairs

Prepared For

Thermo Fisher Scientific Inc.©

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940906



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Reason for revision Initial Release.

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### Disclaimer

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**End of Safety Data Sheet**